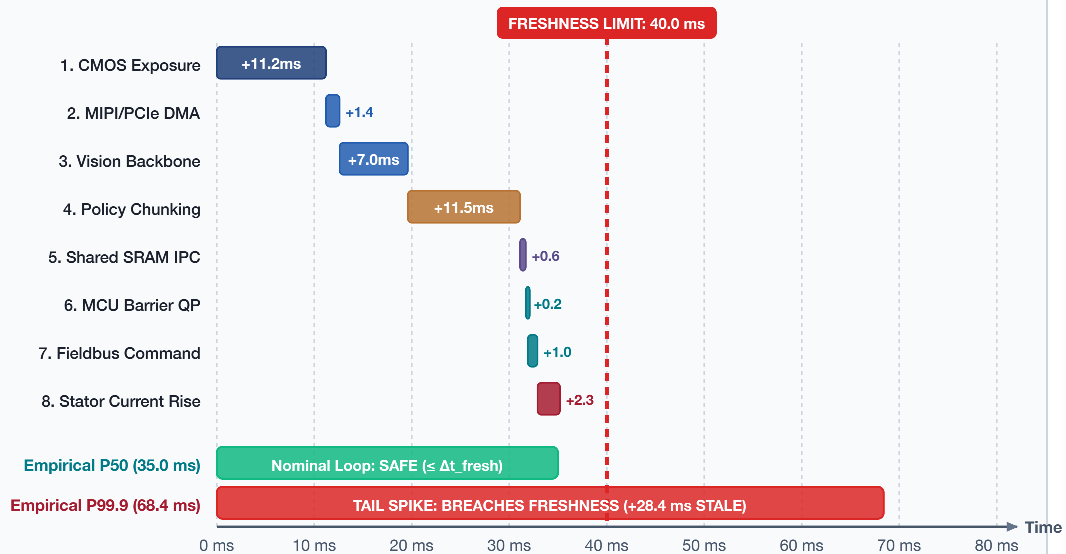
END-TO-END SENSE-TO-ACTUATION LATENCY WATERFALL & FRESHNESS BUDGET

Cumulative Wall-Clock Decomposition: From Photon Transduction to Torque Generation vs. the Freshness Deadline

Stage	Component	Description	P50 Median	P99 Tail
1. TRANSDUCTION	CMOS Exposure	Photons striking well; mid-exposure timestamp	11.2 ms	16.6 ms
2. READOUT/DMA	MIPI & PCIe DMA	Zero-copy ring buffer; crossbar DMA ingest	1.4 ms	3.2 ms
3. TOKENIZER	Vision Backbone	ViT patch embedding; spatial token tensor	7.0 ms	10.5 ms
4. POLICY VLA	Action Chunking	Diffusion / ACT model; 16-step trajectory spline	11.5 ms	13.6 ms
5. MEMORY IPC	Shared SRAM TCM	Lock-free seqlock; MPU-to-MCU mailbox	0.6 ms	2.1 ms
6. REAL-TIME QP	1 kHz MCU Filter	Active-set CBF QP; stopping envelope check	0.2 ms	0.4 ms
7. FIELDBUS	EtherCAT Frame	Cyclic sync torque; PWM gate driver enable	1.0 ms	1.0 ms
8. COIL TORQUE	Stator L/R Rise	Current buildup τ_e; rotor torque $J\ddot{\theta} = \tau\_m$	2.3 ms	3.0 ms

CUMULATIVE SENSE-TO-ACTUATION DELAY WATERFALL (Δt)

Direct Empirical Sum vs. Measured Tail Convolution on Physical Hardware



PHYSICAL IMPACT: CLEARANCE COLLAPSE

Blind Coasting Travel: $d\_react = v_0 \cdot \Delta t$

Operating velocity $v_0 = 1.2$ m/s; clearance D_clear = 20.0 cm

CASE A: Nominal P50 (Δt = 35.0 ms)

d_react 4.2cm	Braking 7.2cm	Margin +8.6cm ✓

Total d_stop = 11.4 cm ≤ 20.0 cm (Certified Safe Operation)

CASE B: Tail Latency P99.9 (Δt = 68.4 ms)

d_react 8.2cm (+4.0cm!)	Braking 7.2cm	Margin 4.6cm ⚠

Total d_stop = 15.4 cm (Safety buffer eroded by 46.5%)

FIRST-PRINCIPLES SYSTEMS IMPLICATION

- Latency is not an elastic software metric; it directly maps into unguided physical travel.
- P99.9 tail latency dictates true safe speed limits, not average computational throughput (P50).
- Stale observations (Δt > Δt_fresh) inject phase lag that destabilizes high-gain feedback loops.